

Course Outline

1. Course Identity

A. Course as listed in CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	DDA5001
Course title (English)	Machine Learning
Course title (Chinese)	机器学习
Units	3
Language of Instruction	English
Description (English)	This course will cover the fundamental aspects of supervised learning and unsupervised learning. The supervised learning part includes: linear regression, least squares, MLE, training versus testing, overfitting, cross-validation, logistic regression, gradient-based optimization methods, regularization, support vector machine, and kernel methods. The unsupervised learning part includes: PCA and its variants, and clustering. Then, the course will also cover some advanced topics, including stochastic gradient methods, Adam family, and basics of deep learning, transformers, and language models.
Description (Chinese)	本课程主要介绍监督学习和无监督学习的基本内容。监督学习部分包括线性回归、最小二乘法、最大似然估计、训练与测试、过拟合、交叉验证、逻辑回归、基于梯度的优化方法、正则化、支持向量机以及核方法。无监督学习部分包括主成分分析及其变体，以及聚类。此外，课程还将涵盖一系列高级机器学习方法，包括随机梯度方法、Adam 优化器系列、深度学习基础、Transformer 架构以及大模型。

B. Corresponding course in CUHK

Please give details of the *closest* corresponding course in CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course in CUHK, please make multiple copies of the block below.

Course code	
Course title (English)	
Course title (Chinese)	
Units	
Description (English)	
Description (Chinese)	

2. Prerequisites / Co-requisites

Please state prerequisites and co-requisites, in terms of courses in CUHK-Shenzhen* or any other requirements (e.g., having taken certain subjects in high school).

(* Because course codes may not yet be stable, please provide both course code and course title.)

A. Prerequisites

Multivariate calculus, probability and statistics, linear algebra

B. Co-requisites

3. Learning Outcomes

Fundamentals

- 1) The students will study the fundamentals of machine learning including training versus testing, overfitting, linear models, logistic regression, SVMs, deep learning etc.
- 2) The students will study how to design and apply algorithms to solve machine learning problems.
- 3) The students will be well prepared for more advanced study in the area of machine learning and have the foundation to do machine learning research.

Skills

- 1) Machine learning is one of the most important technologies today. The students will get familiar with popular machine learning models and learn how to apply learning techniques to solve machine learning problems arising in applications.
- 2) Through the final project, the students will study how language models work and how to write a technical report.

4. Course syllabus

Overview, basic notions of supervised learning, MLE, least squares, training versus testing, over-fitting, cross-validation, logistic regression, gradient-based learning algorithms, regularization, SVMs, kernel methods, PCA and its variants, matrix factorization, k-means, stochastic gradient method, deep learning, transformers, language models.

5. Assessment Scheme

Component/ method	% weight
Homework	30
Course Project	30
Final Exam	40

6. Grade descriptor

General

Grade	Description
A	Outstanding performance on exams and problem sets with excellent understanding on all of the concepts. Clear understanding on the assumptions, limitations and shortcomings of the studied methods, and the potential fixes.
A-	Generally outstanding performance on exams and problem sets with good understanding on almost all of the concepts. Clear understanding on the assumptions, limitations and shortcomings of the studied methods, and the potential fixes.
B+,B, B-	Substantial performance on exams and problem sets with good understanding on most of the concepts. Good understanding on the assumptions, limitations and shortcomings of the studied methods, and the potential fixes.
C +,C C-	Satisfactory performance on exams and problem sets with good understanding on the majority of the concepts, possibly with a few weaknesses. Good understanding on the majority of the assumptions, limitations and shortcomings of the studied methods, and some of the potential fixes, possibly with a few weakness.
D+, D	Barely satisfactory performance on exams and problem sets with weak understanding on some of the concepts. Poor understanding on the assumptions, limitations and shortcomings of the studied methods, and the potential fixes.
F	Unsatisfactory performance on exams and problem sets with poor understanding on almost all of the concepts. Failure to understand the assumptions, limitations and shortcomings of the studied methods, and the potential fixes.

7. Feedback for evaluation

- 1) Official CTE;
- 2) Feedback to instructor and/or teaching assistant.

8. Reading

Recommended Reading:

1. Probabilistic Machine Learning: An Introduction, 9780262369305, K. Murphy
2. Learning from data, 9781600490064, Abu-Mostafa, Y. S., Magdon-Ismail, M., & Lin, H. T.
3. Deep Learning: Foundations and Concepts, 9783031454677, Bishop C., & Bishop H.

9. Course components

Activity	Hours/week
Homework	2
Lecture	3
Reading and coding	2

10. Indicative teaching plan

Week	Content/ topic/ activity
1	Overview, concepts and feasibility of learning
10	Stochastic gradient methods and Adam family
11-13	Basics of deep learning, transformers, and language models
14	Review and final exam
2	Linear regression – least squares, MLE
3	Training versus testing
4	logistic regression, gradient-based optimization methods
5	gradient-based optimization methods, overfitting
6	Overfitting (validation and regularization)
7	SVM, kernel methods
8-9	Dimensionality reduction, clustering

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections.

[Example: 150 students for lecture (x 2); 30 students for tutorials (x 10)]

200 students for lecture * 2 per week

12. Any other information

13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences?

Have the details (as in this document) been approved at School or other level in CUHK-Shenzhen?

Submit application

14. Version date

Version number	1
As of (date)	2025-07-29